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$$\int_{1}^{16} \frac{1}{\sqrt[4]{x^3}+x}dx \overset{x=u^4}{\Rightarrow} \int \frac{4u^3}{u^4+u^3}du = 4\int \frac{1}{u-1}du \overset{t=u-1}{=} 4\int \frac{1}{t}dt$$

$$= 4\ln|t| + c \overset{t=u-1}{=} 4\ln|u-1| + c \overset{u=\sqrt[4]{x}}{=} 4\ln(\sqrt[4]{x}-1) + c$$

$$\Rightarrow 4\left[\ln(\sqrt[4]{x}+1)\right]_1^{16} = 4(\ln(3) - \ln(2))$$

References